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EXPERIMENT 4

Aim: To Perform A Case Study on cloud for IoT: Microsoft Azure.

Theory:

Case Study: Cloud for IoT – Microsoft Azure

Introduction

The Internet of Things (IoT) is transforming industries by enabling real-time data collection, analytics, and automation. Cloud platforms like Microsoft Azure IoT provide scalable, secure, and intelligent solutions for IoT deployment. This case study explores how Microsoft Azure IoT Hub has been leveraged in real-world applications.

Azure IoT Overview

Microsoft Azure IoT is a suite of cloud-based services that facilitate IoT device connectivity, data processing, and AI-driven insights. Key components include:

- Azure IoT Hub – Central messaging hub for IoT devices
- Azure Digital Twins – Digital modeling of real-world environments
- Azure IoT Edge – Enables edge computing for low-latency processing
- Azure Time Series Insights – Visualization and analysis of IoT data
- Azure Security Center for IoT – Protects connected devices

Case Study: Remote Monitoring in Manufacturing

Company: Rolls-Royce Aerospace

Challenge:

- High maintenance costs for jet engines
- Need for real-time performance monitoring
- Inefficient predictive maintenance models

Solution:

- Rolls-Royce integrated Azure IoT Hub to connect thousands of jet engines.
- Azure Stream Analytics processed engine data in real time.
- Azure Machine Learning developed predictive maintenance models.
- Power BI and Time Series Insights provided visualization for engineers.



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Results:

- 30% reduction in maintenance costs
- Improved flight safety through early fault detection
- Optimized fuel efficiency with real-time analytics

Benefits of Microsoft Azure for IoT

- Scalability: Supports millions of devices with elastic cloud resources.
- Security: End-to-end encryption and compliance with industry standards.
- Integration: Works seamlessly with AI, ML, and edge computing.
- Low Latency: Azure IoT Edge enables real-time processing at the source.

Conclusion: Thus we successfully Performed A Case Study on cloud for IoT: Microsoft Azure.